OLIST BRAZIL | SQL PORTFOLIO PROJECT

Ad-hoc Data Analysis Questions

Brazilian E-Commerce Analytics: Orders, Payments, Freight & Delivery Performance

Purpose
Use SQL to explore the Olist Brazilian e-commerce dataset and answer the business analysis requests about order trends, customer geography, payment value, freight, delivery performance, and payment behavior.

Question Set

The following 17 questions follow the supplied problem statement.

1. Exploratory Data Analysis

Q1. What are the data types of all columns in the **customers** table?

Q2. What is the time range between which the orders were placed?

Q3. Count the cities and states of customers who placed orders during the given period.

2. In-depth Exploration

Q4. Is there a growing trend in the number of orders placed over the past years?

Q5. Is there any monthly seasonality in the number of orders placed?

Q6. During what time of day do Brazilian customers mostly place their orders? Use the categories given in the problem statement:
• 0-6 hours: Dawn
• 7-12 hours: Morning
• 13-18 hours: Afternoon
• 19-23 hours: Night.

3. Evolution of E-commerce Orders in Brazil

Q7. Get the month-on-month number of orders placed in each state.

Q8. How are customers distributed across all states?

4. Impact on the Economy: Payment Value, Price and Freight

Q9. Get the percentage increase in order cost from 2017 to 2018, including only January through August. Use **payment_value** from the payments table to measure order cost.

Q10. Calculate the total and average order-item price for each state.

Q11. Calculate the total and average order freight value for each state.

5. Analysis Based on Sales, Freight and Delivery Time

Q12. In a single query, calculate (a) the number of days taken to deliver each order from purchase to customer delivery, and (b) the difference in days between the estimated and actual delivery dates.
Q13. Find the top five states with the highest and lowest average freight value.
Q14. Find the top five states with the highest and lowest average delivery time.
Q15. Find the top five states where delivery is fastest compared with the estimated delivery date. Use the difference between average actual and estimated delivery dates to compare states.

6. Analysis Based on Payments

Q16. Find the month-on-month number of orders placed using different payment types.
Q17. Find the number of orders placed based on the number of payment installments made.

Scope Note

This document lists the original business analysis questions. The analysis results and SQL queries are documented separately in the project README and SQL file.